

Lab 3 Data Dictionary

1. Customers

Column	Data Type	Description	Allowed Values / Typical Range
customer_id	VARCHAR	Unique customer identifier (e.g. CUST00001).	CUST00001–CUST02000 (plus 5 duplicate records for data cleaning).
customer_name	VARCHAR	Customer company name.	Generated business names.
industry	VARCHAR	Industry sector.	Retail, Manufacturing, Healthcare, Logistics, Financial Services, Education, Energy, Professional Services, Technology, Public Services
customer_segment	VARCHAR	Customer size.	Enterprise, Mid-Market, Small Business (<i>some values intentionally stored in lowercase for cleaning practice</i>)
region	VARCHAR	Geographic sales region.	North Region, South Region, East Region, West Region, Central Region, Global Accounts (<i>some NULL values</i>)
customer_since_date	DATE	Date customer relationship started.	Approximately 2017–2024
credit_rating	VARCHAR	Customer credit rating.	A, B, C, D (<i>some NULL values</i>)
credit_limit_CUR	DECIMAL(18,2)	Approved customer credit limit.	Approximately 10,000–800,000

2. Products

Column	Data Type	Description	Allowed Values / Typical Range
product_id	VARCHAR	Unique product identifier.	PROD0001–PROD0150
product_category	VARCHAR	Product or service category.	Software, Consulting, Support, Hardware, Managed Services, Data Services, Analytics Services, Training Services (some NULL values)
product_name	VARCHAR	Product name.	Generated product names
unit_cost_CUR	DECIMAL(18,2)	Cost per unit.	Approximately 30–1,200
selling_price_CUR	DECIMAL(18,2)	Selling price per unit.	Approximately 60–3,000

Primary Key: product_id

3. Invoices

Column	Data Type	Description	Allowed Values / Typical Range
invoice_id	VARCHAR	Unique invoice identifier.	INV00000001–INV00018000
customer_id	VARCHAR	Customer receiving the invoice.	Existing customer IDs
product_id	VARCHAR	Product or service invoiced.	Existing product IDs
invoice_date	DATE	Invoice creation date.	2024-01-01 to 2025-12-31
due_date	DATE	Payment due date.	Typically 15–60 days after invoice date

quantity	INTEGER	Number of units sold.	Typically 1–320
invoice_amount_CUR	DECIMAL(18,2)	Total invoice value.	Approximately 150–950,000 (<i>one intentional outlier = 9,999,999</i>)
payment_status	VARCHAR	Current payment status.	Paid, Partially Paid, Unpaid, Overdue, Written Off, Overpaid

Primary Key: invoice_id

Foreign Keys

- product_id → products.product_id

4. Payments

Column	Data Type	Description	Allowed Values / Typical Range
payment_id	VARCHAR	Unique payment identifier.	PAY00000001–PAY00025000
invoice_id	VARCHAR	Related invoice.	Existing invoice IDs
payment_date	DATE	Date payment received.	Between invoice due date and 2026-01-31
payment_amount_CUR	DECIMAL(18,2)	Amount received.	Positive monetary values (may be partial or exceed invoice value for overpayments)
payment_method	VARCHAR	Payment method used.	Bank Transfer, Card, Digital Payment, Direct Debit (<i>some NULL values</i>)

Primary Key: payment_id

Foreign Key

- invoice_id → invoices.invoice_id

5. Transactions

Column	Data Type	Description	Allowed Values / Typical Range
transaction_id	VARCHAR	Unique transaction identifier.	TXN000000001– TXN000100000
transaction_date	DATE	Accounting transaction date.	2024-01-01 to 2026-01-31
customer_id	VARCHAR	Related customer.	Existing customer IDs
invoice_id	VARCHAR	Related invoice.	Existing invoice IDs
transaction_type	VARCHAR	Accounting transaction type.	Invoice, Payment Summary, Credit Note, Adjustment, Write-off (<i>some NULL values</i>)
amount_CUR	DECIMAL(18,2)	Transaction amount.	Positive or negative monetary values

Primary Key: transaction_id

Foreign Key

- invoice_id → invoices.invoice_id